Q2 — Design Work

Since the midterm, I have completed three additional visualisations and refined the existing code through iterative design, testing, and debugging.

Design Iterations and Technical Challenges

The Heart Rate vs Pace scatter plot required implementing Pearson's correlation coefficient and least-squares linear regression from mathematical formulae. The initial version displayed only data points, making trends invisible. I added a regression line and correlation value ($r = -0.67$), revealing that lower heart rates correlate with faster paces. A key challenge was parsing pace data from "6:30" string format into decimal minutes (6.5), which I debugged using Chrome DevTools breakpoints to trace the data flow through `paceToMinutes()`. After the midterm, I further refined the scatter plot by mapping point size to running distance, creating a bubble chart that communicates three data dimensions simultaneously.

For Pace Progress, raw pace values produced a noisy graph. I implemented a trailing moving average (window = 10 runs) to smooth fluctuations while preserving the downward trend. I also added colour-coded pace zone backgrounds (Fast, Moderate, Easy) to provide immediate visual context for each data point.

The Goal Tracker uses a semicircular gauge rather than a bar chart, because a gauge better communicates percentage progress toward a target. The implementation required trigonometric calculations (cos/sin) for polar coordinate positioning along a 270-degree arc.

Code Quality and Testing

Reviewer feedback identified three issues I addressed: I extracted tooltip rendering into a reusable `Tooltip` class (separation of concerns), fixed implicit global variables from missing `let` declarations, and consolidated duplicate `paceToMinutes()` into `helper-functions.js` (DRY principle). All visualisations use asynchronous callbacks in `loadTable()` with a loaded flag, ensuring the UI remains responsive during CSV loading.

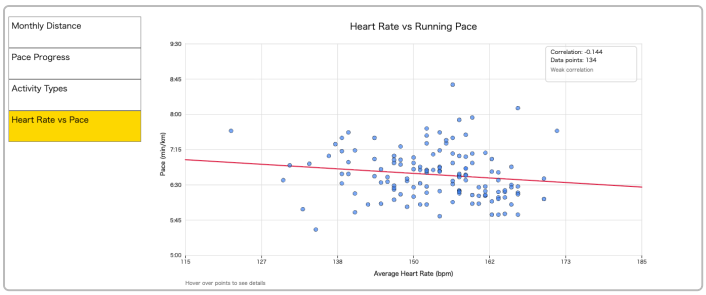
I wrote unit tests (`tests.js`) for key functions — validating `paceToMinutes()`, `formatPace()`, Pearson's correlation, linear regression, moving average, and colour mapping — using black-box testing with known inputs and expected outputs. I also created system test cases to verify each visualisation's rendering, interaction, and data integrity across all seven extensions.

New Visualisations

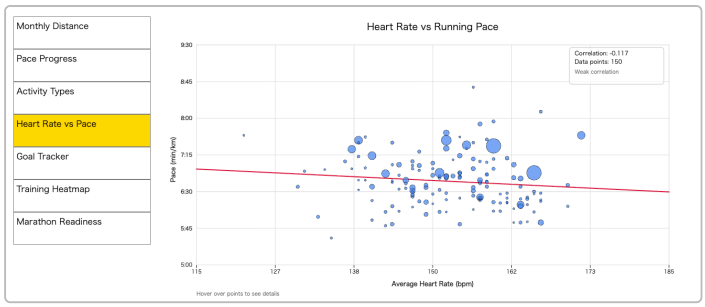
I developed two novel chart types: a Training Heatmap adapting GitHub's contribution graph to display daily distances on a calendar grid (nested loops, colour mapping), and a Marathon Readiness Radar plotting five normalised metrics on a spider chart (trigonometry at 72-degree intervals). Both were built from scratch without external libraries.

These additions transform the project from individual metric displays into an integrated analytical dashboard for marathon preparation.

Figure 1: Heart Rate vs Pace — Design Iteration



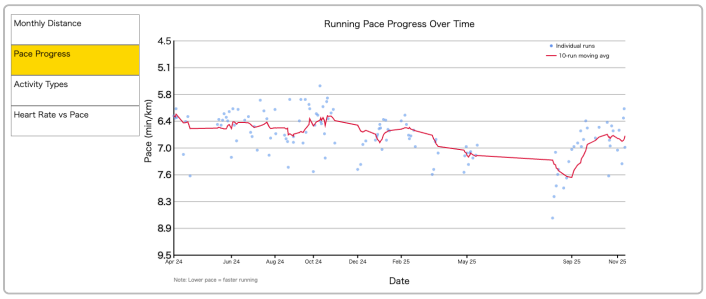
Midterm: uniform point sizes



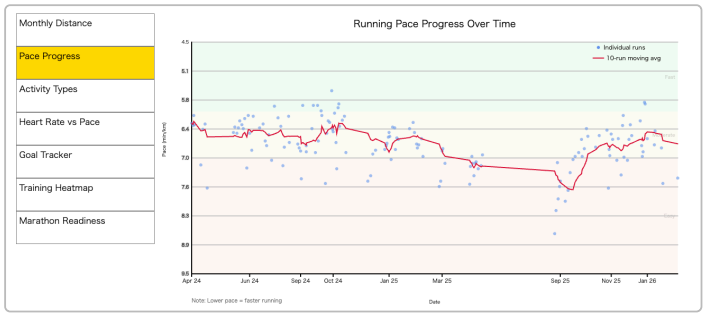
Final: bubble chart with distance-proportional sizing

> Point size now encodes running distance, adding a third data dimension. Longer runs appear as larger bubbles.

Figure 2: Pace Progress — Design Iteration



Midterm: plain background



Final: colour-coded pace zone backgrounds

> Pace zone backgrounds provide immediate visual context, helping identify which training zone each run falls within.